

Cognitive Process Theory of Writing: Flower and Hayes

1 Origins

The Cognitive Process Theory of Writing emerged from the collaborative research of Linda Flower, a rhetorician, and John R. Hayes, a cognitive psychologist, both working at Carnegie Mellon University in the late 1970s and early 1980s. Their partnership was notable for its interdisciplinary character: Flower brought expertise in rhetoric and composition pedagogy, while Hayes contributed the methodological rigor of cognitive psychology and protocol analysis. Together, they sought to move the study of writing beyond the prescriptive stage models that had dominated composition instruction—models that presented writing as a linear sequence of prewriting, writing, and rewriting—and toward an empirically grounded account of what writers actually do when they compose.

Their landmark paper, “A Cognitive Process Theory of Writing,” was published in *College Composition and Communication* (Volume 32, Number 4) in December 1981. This article has since become one of the most cited works in composition studies and is widely regarded as a foundational text in the field. The paper presented a theory of the cognitive processes involved in composing, drawing on data gathered through think-aloud protocol analysis, a method borrowed from cognitive psychology in which writers verbalize their thoughts while composing. By asking participants to say aloud everything that passed through their minds during a writing task, Flower and Hayes were able to construct detailed maps of the mental operations underlying composition, revealing the recursive, non-linear nature of writing processes that had been obscured by earlier stage models.

The intellectual context in which Flower and Hayes worked was shaped by the broader “process movement” in composition studies, which had been gaining momentum since the 1960s. Scholars such as Janet Emig, whose 1971 study *The Composing Processes of Twelfth Graders* employed a case-study approach, and Donald Murray, who championed writing as discovery, had already begun shifting attention from written products to the processes that generated them. Flower and Hayes built upon this trajectory but distinguished their work through its commitment to cognitive science and empirical methodology, offering a model that aspired to systematic, replicable rigor

rather than impressionistic observation.

2 The Model

The Flower and Hayes cognitive process model identifies three major recursive components of writing, embedded within a broader structure that also accounts for the task environment and the writer's long-term memory.

2.1 Planning

Planning encompasses three sub-processes. *Generating* involves retrieving relevant information from long-term memory—facts, ideas, associations, and prior knowledge related to the topic and audience. *Organizing* is the process by which the writer imposes structure on the generated material, selecting and arranging ideas into a coherent plan. *Goal-setting* involves establishing both content goals (what the writer wants to say) and process goals (how the writer intends to proceed), as well as rhetorical goals related to audience, purpose, and persona. Flower and Hayes found that expert writers develop elaborate, hierarchical goal structures, frequently setting and revising sub-goals throughout composing, while novice writers tend to operate with simpler, more static goal representations.

2.2 Translating

Translating is the process of converting the plans, ideas, and mental representations developed during planning into actual written language. This process is cognitively demanding because it requires the writer to juggle multiple constraints simultaneously: the semantic content of what is being expressed, the syntactic demands of sentence construction, the requirements of genre and register, and the physical act of inscription itself. Flower and Hayes emphasized that translating is not merely a transcription of fully formed thought but an active, generative process in which meaning is shaped and sometimes discovered through the act of putting words on the page.

2.3 Reviewing

Reviewing consists of two sub-processes: *evaluating* and *revising*. Evaluating involves reading and assessing what has been written against the writer's goals and intentions, while revising involves making changes to the text or to the writer's plans. Reviewing can occur at any point during composing—not only after a draft is completed—and can lead the writer to cycle back through

planning and translating processes. This recursive quality is one of the model's most important features, distinguishing it from the linear stage models that Flower and Hayes explicitly challenged.

2.4 The Monitor

The model includes a “monitor” function that serves as a metacognitive control mechanism. The monitor determines when the writer switches between processes—for example, deciding when to stop generating ideas and begin translating, or when to pause translating and return to planning or reviewing. The monitor operates according to the writer's habitual strategies and current goals, and its functioning varies significantly between individuals. Skilled writers tend to deploy the monitor more flexibly, moving among processes in response to the demands of the evolving text, while less experienced writers may follow more rigid, habitual patterns.

2.5 Task Environment

The task environment encompasses all external factors that influence the writing process. These include the rhetorical problem itself (the topic, the audience, the exigency or motivation for writing), any instructions or constraints imposed by the assignment, and the text produced so far. As the writer generates text, the existing text becomes part of the task environment, shaping subsequent planning, translating, and reviewing decisions. This feedback loop between the writer's processes and the emerging text is a critical feature of the model, capturing the way writing is both goal-directed and responsive to its own unfolding.

2.6 Long-Term Memory

Long-term memory stores the writer's knowledge of the topic, knowledge of the audience, and knowledge of writing conventions and genre expectations. This stored knowledge is drawn upon during all phases of composing, particularly during generating and goal-setting. The model recognizes that the accessibility and organization of knowledge in long-term memory significantly influence the quality and character of the writing produced. Expert writers possess not only more extensive knowledge but also more richly organized and readily retrievable knowledge structures.

2.7 Non-Linear, Hierarchical Organization

Perhaps the most consequential feature of the Flower and Hayes model is its insistence that writing processes are not linear but hierarchically organized and recursively deployed. Any process can be embedded within any other process: a writer might begin reviewing, which triggers a new planning

episode, which in turn requires translating. This hierarchical embedding means that writing does not proceed in a fixed sequence but rather through a dynamic interplay of processes, governed by the writer's goals and the demands of the rhetorical situation. The model was represented visually as a flow chart of boxes and arrows, indicating the flow of information among processes—a deliberate departure from the sequential stage diagrams that preceded it.

3 Key Innovations

3.1 Writing as Problem-Solving

One of the most significant conceptual contributions of the Flower and Hayes model was its framing of writing as a form of cognitive problem-solving. Rather than treating writing as the expression of preformed thought, they argued that writers confront a complex rhetorical problem defined by the interaction of topic, audience, purpose, and the writer's own goals. Composing, in this view, involves constructing a representation of the problem, generating and testing potential solutions, and revising both the problem representation and the solution as the writing unfolds. This problem-solving framework connected composition studies to the broader tradition of cognitive science research on expertise and decision-making.

3.2 Expert versus Novice Writer Differences

Flower and Hayes's research, and the extensive body of work it inspired, documented systematic differences between expert and novice writers. Expert writers were found to develop more elaborate and abstract representations of the rhetorical problem, to set and revise goals at multiple levels of abstraction, and to engage in more extensive planning and revising. Novice writers, by contrast, tended to generate content directly from memory with minimal transformation, to devote less attention to audience and rhetorical purpose, and to focus their revising efforts on surface-level corrections rather than global restructuring. Expert writers also produced text in larger sentence-level units and spent considerably more time revising, attending particularly to global problems such as re-sequencing, restructuring, and rewriting large sections of text.

3.3 The Role of Goals and Sub-Goals

The model highlighted the centrality of goal-setting in skilled writing. Flower and Hayes identified four categories of goals that effective writers manage: the effect the writing should have on the reader, the persona the writer wishes to project, how the reader will develop meaning through the text, and the conventional features of the genre. Skilled writers break these high-level goals into

hierarchically organized sub-goals and continually revise them as composing proceeds. Novice writers, by contrast, tend to maintain static, surface-level goals—often limited to fulfilling assignment requirements such as page length—and rarely move beyond their initial representation of the task.

3.4 Knowledge-Telling versus Knowledge-Transforming

Carl Bereiter and Marlene Scardamalia (1987) extended the Flower and Hayes framework with their influential distinction between “knowledge-telling” and “knowledge-transforming” strategies. In knowledge-telling, the writer retrieves information from long-term memory in response to topic and genre cues and transcribes it more or less directly, with minimal cognitive transformation. This strategy is characteristic of novice writers and involves relatively little planning, goal-setting, or revision. In knowledge-transforming, the writer actively works to resolve conflicts between content knowledge and rhetorical goals, engaging in a dialectical process between “content space” and “rhetorical problem space” that generates new ideas, deeper understanding, and more sophisticated text. Bereiter and Scardamalia stressed that knowledge-transforming is not simply an evolved version of knowledge-telling but represents a qualitatively different approach to the writing task. This distinction has had profound implications for writing pedagogy, suggesting that the development of writing expertise involves not merely the accumulation of skills but a fundamental restructuring of how the writer conceptualizes and engages with the composing task.

4 Sondra Perl’s Contribution

Sondra Perl’s 1980 article “Understanding Composing,” published in *College Composition and Communication*, offered a complementary perspective to the Flower and Hayes model that enriched the field’s understanding of the composing process. While Flower and Hayes focused primarily on cognitive operations amenable to verbal protocol analysis, Perl drew attention to the embodied, pre-verbal dimensions of writing.

4.1 Felt Sense

Central to Perl’s contribution is the concept of “felt sense,” a term she borrowed from the philosopher and psychologist Eugene Gendlin. Felt sense refers to the vague, bodily experienced awareness that writers have of their topic before and during composing—a pre-verbal, holistic sense of what one wants to say that is “anchored in the writer’s body.” When a writer is given a topic, the topic evokes images, words, ideas, and diffuse feelings that the writer must attend to and gradually articulate. Perl argued that skilled composing involves a recursive movement between this felt

sense and the emerging text, with writers repeatedly returning to their bodily awareness of meaning in order to generate and evaluate language.

4.2 Retrospective Structuring

Perl coined the term “retrospective structuring” to describe the process by which writers attend to their felt sense and write out of that experiential ground. In retrospective structuring, the writer begins with a feeling or intuition that already exists and works to translate it into structured language. This process is inherently recursive: the writer moves forward by going backward, returning to the felt sense, rereading what has been written, and revisiting the topic or keywords in order to generate new text. Perl’s emphasis on recursion complemented the Flower and Hayes model’s insistence on the non-linear character of composing, while adding an affective and phenomenological dimension that their more strictly cognitive framework did not fully address.

4.3 Writing as Recursive

Perl explicitly rejected the linear plan-write-revise scheme that had governed earlier models of composition. Her observational studies of both basic and experienced writers revealed that composing occurs as a fundamentally recursive process in which writers continually loop back through earlier phases: rereading portions of the text already written, returning to the topic or assignment, and re-engaging with the felt sense. This recursive understanding of composing, shared with Flower and Hayes though arrived at through different methodological and theoretical commitments, became one of the defining insights of the process movement.

5 Methodological Approach

5.1 Think-Aloud Protocols

The methodological signature of the Flower and Hayes research program was the think-aloud protocol, a technique adapted from cognitive psychology. In this method, writers are asked to verbalize everything passing through their minds while composing—not to explain or justify their actions, but simply to give voice to their ongoing thoughts. The resulting verbal protocols are recorded, transcribed, and coded according to a system of categories corresponding to the theorized cognitive processes (generating, organizing, goal-setting, translating, evaluating, revising, and monitoring). Protocol analysis allowed Flower and Hayes to construct detailed, moment-by-moment maps of the cognitive activity involved in writing, revealing patterns that were invisible to external observation of the finished text alone.

5.2 Empirical and Scientific Orientation

Flower and Hayes positioned their work explicitly within the tradition of empirical science, seeking to develop testable models of writing processes grounded in observable data. This orientation distinguished their approach from the more humanistic, interpretive traditions that had dominated composition studies. Their use of protocol analysis, borrowed from the information-processing paradigm in cognitive psychology pioneered by Allen Newell and Herbert Simon, lent their research a methodological rigor that was relatively novel in the study of writing. The resulting model was presented not as a philosophical argument or a pedagogical prescription but as a scientific hypothesis subject to revision in light of new evidence.

5.3 Limitations of the Method

Think-aloud protocols, while powerful, have recognized limitations. The requirement to verbalize may alter the very processes under study, either by diverting cognitive resources from composing or by prompting the writer to engage in processes (such as reflection or evaluation) that they might not otherwise perform. Not all cognitive processes are accessible to conscious introspection, and the method necessarily captures only those aspects of composing that can be verbalized. Moreover, the coding of protocols requires interpretive judgments that introduce an element of subjectivity into what is presented as an objective method. These methodological concerns became focal points for critics of the cognitive process approach.

6 Criticisms

6.1 Post-Process Critique

By the late 1980s and 1990s, a number of scholars began to challenge the cognitive process model on the grounds that it was too universalizing—that it posited a single, generalizable model of writing processes that failed to account for the diversity of writers, contexts, and purposes. Post-process theorists such as Thomas Kent argued that writing is a fundamentally indeterminate, situated, and interpretive activity that cannot be captured by any systematic model. From this perspective, the Flower and Hayes model, with its flow charts and taxonomies, represented an attempt to reduce the irreducible complexity of composing to a set of predictable cognitive operations.

6.2 Social Constructionist Critique

Perhaps the most sustained critique came from social constructionists, who argued that the Flower and Hayes model isolated the individual writer from the social, cultural, and institutional contexts in which writing occurs. Patricia Bizzell, in her influential essay “Cognition, Convention, and Certainty: What We Need to Know About Writing” (1982), argued that the cognitive process model failed to account for the role of discourse communities, social conventions, and ideological forces in shaping both what writers write and how they write it. Bizzell contended that the model’s focus on inner cognitive operations obscured the degree to which writing is a social act, structured by shared conventions and power relations rather than by individual mental processes alone. Lester Faigley and James Berlin similarly critiqued the model’s ideological assumptions, arguing that the problem-solving framework implicitly endorsed a particular vision of the autonomous, rational, goal-directed individual that was itself a culturally specific construction rather than a universal description of human cognition.

6.3 Feminist Critiques

Feminist scholars raised related concerns about the model’s conception of the writing subject. The Flower and Hayes model was accused of presupposing an autonomous, rational agent who composes through deliberate, goal-directed problem-solving—a figure drawn from the tradition of Western liberal individualism that feminist theorists had long critiqued as implicitly masculine. By centering individual cognition and strategic control, the model marginalized the relational, embodied, and affective dimensions of writing that scholars such as Sondra Perl had begun to illuminate. The model was further criticized for its scientism—its adoption of the methods and assumptions of cognitive psychology in a domain that, critics argued, required hermeneutic and critical rather than positivist approaches.

6.4 Flower’s Later Revisions

Linda Flower herself responded to these critiques by substantially revising her theoretical framework. In her 1994 book *The Construction of Negotiated Meaning: A Social Cognitive Theory of Writing*, Flower redefined writing as an interactive social and cognitive process in which meaning is negotiated among competing forces: social expectations, discourse conventions, and the writer’s personal goals and knowledge become “inner voices” whose tensions shape the composing process. Conflict, rather than smooth problem-solving, became central to this revised theory. Flower argued that writers sometimes rise to the “active negotiation of meaning,” creating text in the interplay of alternatives, opportunities, and constraints rather than through the straightforward

execution of plans.

John R. Hayes also published a substantially revised framework in 1996, “A New Framework for Understanding Cognition and Affect in Writing,” which expanded the original model to include motivation (goals, predispositions, beliefs, and cost-benefit estimates), affect, working memory, and a more explicit account of the social and physical dimensions of the task environment. The revised model acknowledged that social elements—audience, collaborators, cultural norms—interact bidirectionally with individual cognitive processes, affective states, and knowledge resources, thereby addressing some of the most pointed criticisms of the 1981 formulation.

7 Legacy

7.1 Influence on Writing Instruction

The Flower and Hayes model exerted an enormous influence on writing pedagogy from the 1980s onward. By demonstrating that writing is a complex, recursive process rather than a linear sequence, the model provided theoretical grounding for process-based pedagogies that emphasize invention, drafting, revision, and peer feedback rather than the evaluation of finished products. The finding that expert and novice writers differ not in the quality of their prose alone but in the sophistication of their planning and revising strategies suggested that these higher-order processes could be taught, shifting pedagogical attention from surface correctness to strategic composing behavior.

7.2 Writing Across the Curriculum

The cognitive process model also influenced the Writing Across the Curriculum (WAC) movement, which emerged in the 1970s and expanded significantly through the 1980s and 1990s. WAC programs, grounded in the principle that writing is integral to learning and thinking in all disciplines, drew on the Flower and Hayes insight that writing is a mode of cognitive processing—not merely a medium for recording pre-existing knowledge, but a process through which knowledge is generated, organized, and transformed. The Bereiter and Scardamalia distinction between knowledge-telling and knowledge-transforming became particularly influential in WAC contexts, providing a theoretical rationale for using writing assignments as tools for deepening students’ understanding of disciplinary content.

7.3 The Process Movement

More broadly, the Flower and Hayes model, together with the work of scholars such as Perl, Emig, Murray, and Elbow, helped to consolidate the process movement as the dominant paradigm in composition studies for over two decades. While the field has since moved through post-process and post-critical phases, the basic insights of the cognitive process theory—that writing is recursive, goal-directed, cognitively complex, and amenable to empirical study—remain foundational assumptions of composition research and pedagogy. The model’s emphasis on the writer’s mental processes opened an enduring line of inquiry into the cognitive, metacognitive, and self-regulatory dimensions of writing that continues to inform contemporary research in educational psychology, cognitive science, and writing studies.

7.4 Enduring Significance

The cognitive process theory’s greatest legacy may be methodological and conceptual rather than strictly theoretical. By insisting that writing could be studied empirically and that its processes could be modeled with the tools of cognitive science, Flower and Hayes transformed composition studies from a largely impressionistic, pedagogical enterprise into a research discipline with aspirations to systematic inquiry. Even scholars who reject the model’s specific claims or its underlying assumptions about cognition and agency work within an intellectual landscape that the cognitive process theory fundamentally shaped. The ongoing dialogue between cognitive and social perspectives on writing—a dialogue inaugurated in large measure by the critiques of Flower and Hayes—remains one of the most productive tensions in the field.

References

- Bereiter, C., & Scardamalia, M. (1987). Knowledge telling and knowledge transforming in written composition. In S. Rosenberg (Ed.), *Advances in applied psycholinguistics, Volume 2: Reading, writing, and language learning* (pp. 142–175). Cambridge University Press.
- Bizzell, P. (1982). Cognition, convention, and certainty: What we need to know about writing. *PRE/TEXT*, 3(3), 213–243.
- Emig, J. (1971). *The composing processes of twelfth graders*. NCTE Research Report No. 13. National Council of Teachers of English.
- Emig, J. (1977). Writing as a mode of learning. *College Composition and Communication*, 28(2), 122–128.
- Flower, L. (1994). *The construction of negotiated meaning: A social cognitive theory of writing*. Southern Illinois University Press.

Flower, L., & Hayes, J. R. (1981). A cognitive process theory of writing. *College Composition and Communication*, 32(4), 365–387.

Hayes, J. R. (1996). A new framework for understanding cognition and affect in writing. In C. M. Levy & S. Ransdell (Eds.), *The science of writing: Theories, methods, individual differences, and applications* (pp. 1–27). Lawrence Erlbaum Associates.

Kent, T. (Ed.). (1999). *Post-process theory: Beyond the writing-process paradigm*. Southern Illinois University Press.

Perl, S. (1980). Understanding composing. *College Composition and Communication*, 31(4), 363–369.